A SEQUENCE OF POINT BLOWUPS WITH CONSTANT RESIDUE FIELD AND A FIXED EXCEPTIONAL EQUATION COMES FROM A NONSINGULAR ARC

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the power estimate for the distinguished element against the powers of the maximal ideals, the rewriting of generators of the successive maximal ideals and the induction computing the lengths of the successive quotients, and the limit construction recognizing a complete discrete valuation ring together with the lifting of the map along the blowups.

Editorial note. The label lemma-sequence-blowups names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-sequence-blowups –start 2802 –end 2891 –local-id 28074 returned RESULT=OK (L28074 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Blowing up along a regular element and the ideal of the exceptional divisor. Standard description of a blowup: the ideal of the exceptional divisor pulls back to the ideal generated by the blown up element, and multiplicities of powers translate into membership in powers of that ideal. Supplies the claim that a power of the distinguished element is not in the corresponding power of the maximal ideal of the local ring at the next step.
- Characterization of affine blowup algebras. Divisors, the lemma characterizing an affine blowup algebra, used as one of the two ways of lifting the local homomorphism to the blowup. Supplies the explicit construction identifying the local rings of the blowups with localizations of repeated affine blowup algebras, hence the lift of the map to the limit.
- Local rings of a blowup at a closed point and residue fields. Standard facts about the local ring of a blowup at a point and the equality of residue fields along the sequence. Supply the identification of residue fields, which allows coefficients to be chosen in the original local ring.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BFL: sources/stacks-chow/tags.txt, source lines 14389–14389. Prerequisite pointer invoked at proof line(s) 2886; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let*

$$(X, p) = (X_0, p_0) \leftarrow (X_1, p_1) \leftarrow (X_2, p_2) \leftarrow (X_3, p_3) \leftarrow \ldots$$

be a sequence of blowups such that

(1) *p_i is closed, maps to p_{i-1}, and $\kappa(p_i) = \kappa(p_{i-1})$,*
(2) *there exists an $x_1 \in \mathfrak{m}_p$ whose image in $\mathfrak{m}_{p_i}$, $i > 0$ defines the exceptional divisor $E_i \subset X_i$.*

Then the sequence is obtained from a nonsingular arc $a : T \to X$ as above.

Proof. Let us write $\mathcal{O}_n = \mathcal{O}_{X_n, p_n}$ and $\mathcal{O} = \mathcal{O}_{X,p}$. Denote $\mathfrak{m} \subset \mathcal{O}$ and $\mathfrak{m}_n \subset \mathcal{O}_n$ the maximal ideals.

We claim that $x_1^t \notin \mathfrak{m}_n^{t+1}$. Namely, if this were the case, then in the local ring $\mathcal{O}_{n+1}$ the element x_1^t would be in the ideal of $(t+1)E_{n+1}$. This contradicts the assumption that x_1 defines E_{n+1}.

For every n choose generators $y_{n,1}, \ldots, y_{n,t_n}$ for $\mathfrak{m}_n$. As $\mathfrak{m}_n\mathcal{O}_{n+1} = x_1\mathcal{O}_{n+1}$ by assumption (2), we can write $y_{n,i} = a_{n,i}x_1$ for some $a_{n,i} \in \mathcal{O}_{n+1}$. Since the map $\mathcal{O}_n \to \mathcal{O}_{n+1}$ defines an isomorphism on residue fields by (1) we can choose $c_{n,i} \in \mathcal{O}_n$ having the same residue class as $a_{n,i}$. Then we see that

$$\mathfrak{m}_n = (x_1, z_{n,1}, \ldots, z_{n,t_n}), \quad z_{n,i} = y_{n,i} - c_{n,i}x_1$$

and the elements $z_{n,i}$ map to elements of $\mathfrak{m}_{n+1}^2$ in $\mathcal{O}_{n+1}$.

Let us consider

$$J_n = \mathrm{Ker}(\mathcal{O} \to \mathcal{O}_n/\mathfrak{m}_n^{n+1})$$

We claim that $\mathcal{O}/J_n$ has length $n+1$ and that $\mathcal{O}/(x_1) + J_n$ equals the residue field. For $n = 0$ this is immediate. Assume the statement holds for n. Let $f \in J_n$. Then in $\mathcal{O}_n$ we have

$$f = ax_1^{n+1} + x_1^n A_1(z_{n,i}) + x_1^{n-1}A_2(z_{n,i}) + \ldots + A_{n+1}(z_{n,i})$$

for some $a \in \mathcal{O}_n$ and some A_i homogeneous of degree i with coefficients in $\mathcal{O}_n$. Since $\mathcal{O} \to \mathcal{O}_n$ identifies residue fields, we may choose $a \in \mathcal{O}$ (argue as in the construction of $z_{n,i}$ above). Taking the image in $\mathcal{O}_{n+1}$ we see that f and ax_1^{n+1} have the same image modulo $\mathfrak{m}_{n+1}^{n+2}$. Since $x_n^{n+1} \notin \mathfrak{m}_{n+1}^{n+2}$ it follows that J_n/J_{n+1} has length 1 and the claim is true.

Consider $R = \lim \mathcal{O}/J_n$. This is a quotient of the $\mathfrak{m}$-adic completion of $\mathcal{O}$ hence it is a complete Noetherian local ring. On the other hand, it is not finite length and x_1 generates the maximal ideal. Thus R is a complete discrete valuation ring. The map $\mathcal{O} \to R$ lifts to a local homomorphism $\mathcal{O}_n \to R$ for every n. There are two ways to show this: (1) for every n one can use a similar procedure to construct $\mathcal{O}_n \to R_n$ and then one can show that $\mathcal{O} \to \mathcal{O}_n \to R_n$ factors through an isomorphism $R \to R_n$, or (2) one can use Divisors, Lemma 0BFL to show that $\mathcal{O}_n$ is a localization of a repeated affine blowup algebra to explicitly construct a map $\mathcal{O}_n \to R$. Having said this it is clear that our sequence of blowups comes from the nonsingular arc $a : T = \mathrm{Spec}(R) \to X$. $\square$